

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Two-axis SAS	105-87003	7.5	3300	24750
Two-axis SAS	D134-891	7.5	3300	24750
Two-axis SAS	DSK0-90008	7.5	3300	24750

7. SYSTEM DESCRIPTION

The two-axis Stability Augmentation System (SAS) is a “hands-on” system designed to improve dynamic stability of the helicopter and ease pilot workload by minimizing the helicopter reaction to external disturbances such as turbulence.

The SAS comprises two independent subsystems, roll and pitch, having a common control panel. Each subsystem comprises an electromechanical linear actuator controlled by a gyro-amplifier.

A controller mounted on the LH fuselage interior houses both gyro-amplifiers and a static inverter supplying 115V AC to each gyro-amplifier and actuator.

The gyro-amplifier senses rotation rates of the helicopter and sends position signals to the actuator. The actuator, installed in series of the respective cyclic control linkage immediately below the hydraulic unit, converts signals from the gyro-amplifier into corresponding control movements by extending or retracting its control rod length. The actuator authority is limited to $\pm 5\%$ of the total respective axis control movement.

The two-axis SAS control panel, located in the center console, has a circuit breaker, a two-position (OFF-STDBY) toggle switch (for energizing the system) and two pushbutton switches (for mode select of each axis) each having two legends (STDBY - yellow and ON - green). An SAS pushbutton at the base of the cyclic stick grip is used for simultaneously switching both axes from the ON mode to the STDBY mode.

Placing the SAS power switch to STDBY energizes the system inverter supplying power to the gyros. When a gyro has reached its operating speed (approx 10 sec), the respective STDBY legend will come on.

NOTE In the standby mode (STDBY legend on), each actuator is locked in a neutral, centered position.

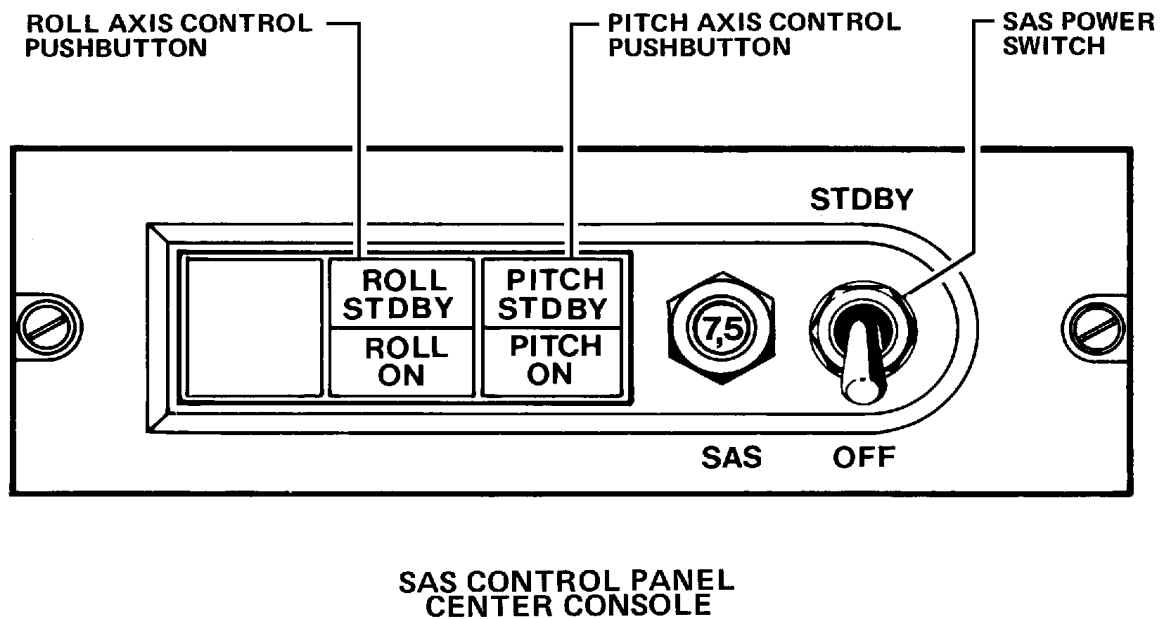
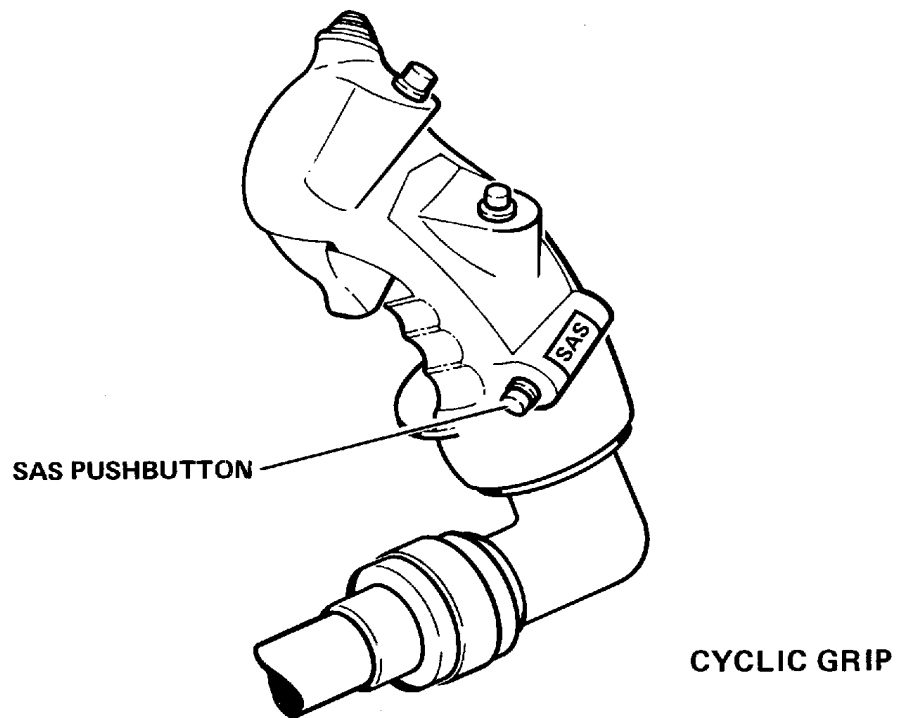


Fig. 1 SAS Controls and Indicators

